

Scientific Reviewer

Strong empirical systems — not universal Chinchilla

Review posture

Accept-level ICLR Oral for real-robot scaling substance in the single-task IL regime. Punish overclaim as universal robot Chinchilla / multi-task VLA law / transferable Kaplan exponents.

Cite Fig.5 + §4.3 recipe + Table 1; caveat §7.

Claim under review

What the paper actually asserts

Primary claim

Data scaling alone can yield single-task imitation policies that deploy zero-shot to any same-category object in any environment — with generalization (optimality gap) following approximate power laws in $\#environments$, $\#objects$, and $\#env-object$ pairs; diversity $\gg$ raw demo count past $K \approx 50$.

>40k demos

collected UMI

>15k rollouts

real Franka

4 tasks

tabletop similar

DP + UMI

one stack

Strengths

Why this clears the Oral bar

Real scale

>15k real-robot rollouts with blind multi-policy eval under identical inits — rare in IL scaling

claims.

Clean OOD ops

Env × object operationalization (not synthetic single-factor); nested subsets; final ckpt

discipline.

Actionable recipe

32 pairs × 50 demos validated on two held-out tasks with afternoon-scale collection.

Honest metrics

MSE failures documented (LoRA trap); prefer human staged scores; 95% CI shaded.

Predictive attempt

Fig.5 fits + 1191-pair extrapolation stated as prediction (left unverified) — falsifiable.

Evidence core — Fig. 5

Six power-law panels · $|r|$ 0.942–0.991 · demo axis fails

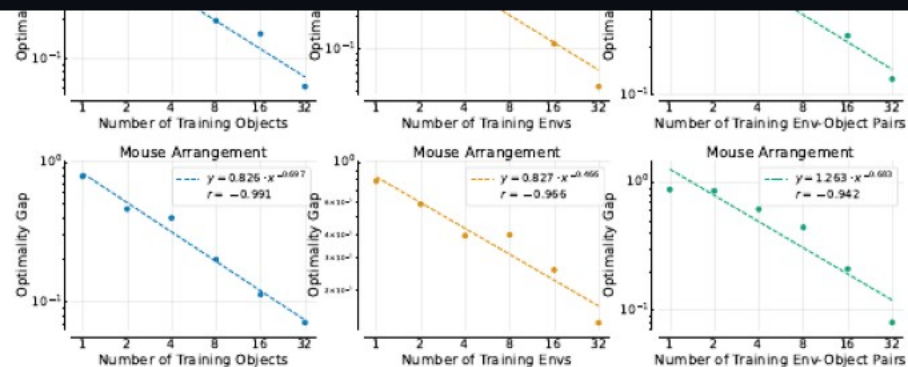


Figure 5: **Power-law relationship.** Dashed lines represent power-law fits, with the equations provided in the legend. All axes are shown on a logarithmic scale. The correlation coefficient (Pearson's r) indicates a power-law relationship between the generalization ability and the number of objects, environments, and environment-object pairs. See Appendix G.1 for data scaling laws on MSE.

ing further emphasizes that expanding the diversity of environments and objects is more effective than merely increasing the number of demonstrations for each individual environment or object.

4.2 POWER-LAW FITTING AND QUANTITATIVE ANALYSIS

We next explore whether our experimental results follow power-law scaling laws, as seen in other domains. Specifically, if two variables Y and X satisfy the relation $Y = \beta \cdot X^{\alpha}$, they exhibit a power-law relationship. Applying a logarithmic transformation to both Y and X reveals a linear relationship: $\log(Y) = \alpha \log(X) + \log(\beta)$. In our context, Y represents the optimality gap, defined as the deviation from the maximum score (i.e., $1 - \text{Normalized Score}$), while X can denote the number of environments, objects, or demonstrations. Using data from our previous experiments with a 100% fraction of demonstrations, we fit a linear model to the log-transformed data, as shown in Fig. 5. Based on all the results, we summarize the following data scaling laws:

¹Although we recognize the irreducible errors Y_{∞} associated with scaling the data alone, fitting a three-parameter model $Y = \beta X^{\alpha} + Y_{\infty}$ does not seem statistically justified given that we have only 6 data points.

Evidence — Table 1 recipe transfer

Cross-task check of 32×50 · SR 85–92.5%

85.0 ± 19.4%

Pour Water

92.5 ± 9.7%

Mouse Arrangement

87.5 ± 17.1%

Fold Towels

90.0 ± 14.1%

Unplug Charger

Weaknesses — threats to validity (1/2)

Scope & statistical limits

Only 4 tasks

Similar tabletop difficulty — dexterous / long-horizon / contact-rich exponents unknown.

6-point fits

No Y_∞ term (authors: statistically unjustified); 1191-pair extrapolation unverified.

Single algorithm

DP + UMI stack confounds “data law” vs “interface / policy family law.”

Human scoring

Blind batches mitigate but do not eliminate subjectivity; batch incomparability (App.E.2).

Weaknesses — threats to validity (2/2)

Missing axes & selection bias

No task / language axis

Cannot adjudicate generalist VLA / multi-task scaling — authors defer to OXE/DROID-scale corpora.

UMI-friendly selection

Texture-poor scenes & large occluders avoided — task-family bias on what “env” means.

Per-env variance

Tab.12: Pour env#2 = 40% SR — mean ~90% overstates robustness without stratification.

Embodiment fixed

Single Franka deploy; collection = human+UMI — no multi-robot law.

CLEAR vs TENTATIVE — citation discipline

What you may claim in the data cluster

CLEAR — cite freely

- Diversity $\gg$ demo depth for single-task IL under DP+UMI
- Recipe $32 \times 50 \rightarrow \sim 90\%$ on paper's 4 tasks
- Fig.5 exponents / Tab.1 / Tab.2 capacity morals
- Env gen harder than object gen (these tasks)

TENTATIVE — punish if overclaimed

- Universal robot Chinchilla / Kaplan exponent transfer
- Multi-task / language / multi-embodiment laws
- 1191-pair Mouse prediction as measured fact
- ACT / π_0 / VLA exponents equal to DP

VERDICT

Strong empirical systems — not universal Chinchilla

ACCEPT — landmark empirical systems

Genuine scaling-law substance for the single-task IL regime with real robots, actionable recipe, and honest metric analysis.

PUNISH: marketing as universal robot Chinchilla; invented multi-task laws; silent DP-only / 4-task / 6-point caveats; claiming 1191 pairs verified.

Citation recipe

Fig.5 + §4.3 recipe + Table 1 · caveat §7 (single-task, IL-only, 4 tasks, DP-only)

Method notes worth probing

What a careful reviews asks next

Nested design

Larger always contain smaller — fair distribution; good. Epochs scaled with size — check undertrain of large M.

Final vs best-val

Final checkpoint used — reduces val leakage; report sensitivity.

Demo fractions

Skip if total demos ≤ 100 — grid holes at tiny N; document.

Capacity split

ViT helps / U-Net doesn't — capacity story incomplete without action-side scalable architecture search.

Constant-total redraw

App.G.2 strengthens diversity claim — should be main-text for causal readers.

Numbers tattoo — pin these

Every deck carries the same spine numbers

diversity $\gg$ depth

envs/objs/pairs $\gg$ K

32 $\times$ 50

pairs $\times$ demos $\rightarrow$ $\sim$ 90%

85-92.5%

Tab.1 SR $\cdot$ 4 tasks

$\alpha \approx -0.47...-0.84$

Fig.5 exponents

0.66 / 0.81 / 0.90

ViT-S / B / L

0.88 / 0.90 / 0.83

U-Net S / base / L

0.00 / 0.03

frozen / LfS fail

K $\approx$ 50

plateau / pair

>40k / >15k

demos / rollouts

1191 pairs

MA score 0.99 extrap.

DP + ACT TE

@5 Hz $\cdot$ horizon 16

FT 0.90 / LoRA 0.72

full FT required

Exam bite — Reviewer

EXAM BITE

Cite Fig.5 + §4.3 recipe + Table 1; caveat §7 (single-task, IL-only, 4 tasks, DP-only).
Strong empirical systems — not universal Chinchilla. 6-point fits; 1191 unverified; no task/lang axis.

Scientific Reviewer — close

Takeaways + exam bite

1. Strengths: >15k real rollouts, blind eval, actionable 32×50 recipe, honest MSE analysis.
2. Weaknesses: 4 tasks, 6-point fits, DP+UMI confound, no task/lang axis, env variance.
3. Verdict: Accept systems+data laws for single-task IL; punish universal Chinchilla overclaim.
4. Tattoo: diversity $\gg$ depth · 32×50→~90% · Fig.5 α · ViT>U-Net · frozen/LfS fail.

EXAM BITE

Cite Fig.5 + recipe + Tab.1; caveat §7 — strong systems, not universal Chinchilla.